

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Civil & Allied Engineering Semester: V
Subject Code & Name: BTCVPE506G/BTCIPE506G & Material, Testing and Evaluation

Max Marks: 60

Date:12/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Classify and Explain 'Physical Properties' of Polymers.	CO 2	6
B) Compare Hard and Soft wood with minimum 10 properties.	CO 4	6
C) Explain with Figure the Variability of Materials.	CO 2	6
Q.2 Solve Any Two of the following.		12
A) Describe in detail 'Aerated Autoclaved Block'.	CO 2	6
B) Write minimum Yield stress and applications for following: a) Fe E 220 b) Fe 410 c) Fe 330	CO 2	6
C) Write note on Engineered wood products.	CO 2	6
Q. 3 Solve Any Two of the following.		12
A) Discuss the properties of Fiber Reinforced Concrete.	CO 2	6
B) Describe in detail the environmental and health impact of typical composite material.	CO 2	6
C) Explain the factors responsible for determining the Choice of Proper Material in construction.	CO 2	6
Q.4 Solve Any Two of the following.		12
A) Describe Photo catalytic admixture in detail.	CO 2	6
B) Report Advantages and Disadvantages of bacterial concrete.	CO 2	6
C) Discuss 'Hempcrete' as material in construction industry.	CO 2	6
Q. 5 Solve Any Two of the following.		12
A) Enlist and Explain material factors that affect visual testing.	CO 2	6
B) Classify Destructive testing and explain any one.	CO 2	6
C) Enlist with minimum six points the necessity of material testing.	CO 2	6

***** End *****